

Individual Assignment 5

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Set up

Packages

```
# general use and cleaning  
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --  
v dplyr      1.2.0      v readr      2.2.0  
v forcats    1.0.1      v stringr    1.6.0  
v ggplot2    4.0.2      v tibble     3.3.1  
v lubridate  1.9.5      v tidyr      1.3.2  
v purrr      1.2.1
```

```
-- Conflicts ----- tidyverse_conflicts() --
```

```
x dplyr::filter() masks stats::filter()
```

```
x dplyr::lag()     masks stats::lag()
```

```
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
library(here)
```

here() starts at /Users/maisieprestridge/Desktop/github/week-05_spring-
2026_aquatic-inverts

```
library(janitor)
```

Attaching package: 'janitor'

The following objects are masked from 'package:stats':

chisq.test, fisher.test

```
library(snakecase)

# wrap axis labels
library(scales)
```

Attaching package: 'scales'

The following object is masked from 'package:purrr':

discard

The following object is masked from 'package:readr':

col_factor

```
# reading .xlsx files
library(readxl)

# visualise missing data
library(naniar)

# arrange plots
library(patchwork)
```

Data

```
# taxonomic information
taxon_list <- read_csv(here("data", "taxon_list.csv"))
```

Rows: 50 Columns: 15

-- Column specification -----

Delimiter: ","

chr (15): verbatim_name, taxonRank, scientificName, kingdom, Phylum, Class, ...

i Use ``spec()`` to retrieve the full column specification for this data.

i Specify the column types or set ``show_col_types = FALSE`` to quiet this message.

```
# aquatic invertebrates
aq_ins <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Aquatic Insects")

# site information
sites <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "sites")

# water quality
water_quality <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Water Quality",
  na = c("", "n/a", "N/A", "over", "173+"))
```

```
# creating new clean object from water quality
water_quality_clean <- water_quality |>
  # clean column names
  clean_names() |>
  # filtering join: only include sites in ncos_sites
  semi_join(sites, by = "site") |>
  # create new column to join with later data frames
  unite("date_site", date, site, remove = TRUE) |>
  # group by date_site column
  group_by(date_site) |>
  # calculate median pH, salinity, DO
  summarize(med_ph = median(p_h, na.rm = TRUE),
    med_sal = median(salinity_ppt, na.rm = TRUE),
    med_do = median(dissolved_oxygen_mg_l, na.rm = TRUE)) |>
  # ungroup data frame
  ungroup() |>
  # filter out salinity outliers
  filter(date_site != "2022-08-12_NVBR")
```

```
# creating new clean object from taxon_list
taxon_list_clean <- taxon_list |>
  # converting taxon name to snake case (no spaces or capital letters,
  # only underscores)
  mutate(verbatim_name = to_any_case(verbatim_name, case = "snake"))
```

```

# creating new clean object from aquatic inverts
aq_ins_clean <- aq_ins |>
  # clean column names
  clean_names() |>
  # filtering join: only include sites occurring in ncos_sites
  semi_join(sites, by = c("site")) |>
  # renaming columns
  rename(date = date_on_vial) |>
  # select columns of interest
  select(date, sample_type, site,
          ostracod,
          copepod,
          hemiptera_corixidae_boatman,
          cladocera,
          nematode,
          diptera_ceratopogonidae,
          annelida_oligochaete,
          diptera_chironomid,
          annelida_polychaete) |>
  # filter to only include "filtered beaker" samples
  filter(sample_type %in% c("FB 250", "FB250")) |>
  # convert the data frame to long format
  pivot_longer(cols = ostracod:annelida_polychaete,
               names_to = "taxon",
               values_to = "abundance") |>
  # group by date, site, taxon
  group_by(date, site, taxon) |>
  # calculate average abundance per liter (sum abundance divided by 7.5 liters)
  summarize(ave_lit = sum(abundance, na.rm = TRUE)/7.5) |>
  # ungroup data frame
  ungroup() |>
  # create a new column called `date_site` to join with water quality
  unite("date_site", date, site, remove = FALSE) |>
  # join with water quality
  left_join(water_quality_clean, by = c("date_site")) |>
  # join with taxon list
  left_join(taxon_list_clean, by = c("taxon" = "verbatim_name")) |>
  # add full names of sites
  mutate(site_full = case_when(
    site == "NVBR" ~ "North of Venoco Bridge",
    site == "NEC" ~ "South of Venoco Bridge",

```

```

site == "NMC" ~ "Main Channel",
site == "NPB" ~ "South of Phelps Bridge",
site == "NPB1" ~ "North of Phelps Bridge",
site == "NPB2" ~ "Phelps Road",
site == "NWP" ~ "West Pond",
site == "NDC" ~ "Devereux Creek"
)) |>
# setting factor levels for site
mutate(site_full = fct_reorder(site_full, med_sal, .fun = "median", .na_rm = TRUE),
       site_full = fct_rev(site_full))

```

`summarise()` has regrouped the output.
i Summaries were computed grouped by date, site, and taxon.
i Output is grouped by date and site.
i Use `summarise(.groups = "drop\_last")` to silence this message.
i Use `summarise(.by = c(date, site, taxon))` for per-operation grouping
(`?dplyr::dplyr\_by`) instead.

1. Visualizing differences across sites in major taxon abundance

a. Plan your figure

x-axis: site_full y-axis: log_ave_lit geometry to represent average abundance/L: geom_jitter()
geometry to represent medians: geom_point()

b. Wrangle your data

```

# new object created from clean aquatic inverts
ostracods <- aq_ins_clean |>
# filter to only include copepods
filter(taxon == "ostracod") |>
# log + 1 to transform mean abundance/liter
mutate(log_ave_lit = log(ave_lit + 1))

slice_sample(ostracods)

```

```

# A tibble: 1 x 24
  date_site      date      site  taxon  ave_lit med_ph med_sal med_do

```

```

      <chr>          <dtm>          <chr> <chr>      <dbl> <dbl> <dbl> <dbl>
1 2021-09-23_M01 2021-09-23 00:00:00 M01 ostrac~      0  8.97    91  5.28
# i 16 more variables: taxonRank <chr>, scientificName <chr>, kingdom <chr>,
#   Phylum <chr>, Class <chr>, Subclass <chr>, Superorder <chr>, Order <chr>,
#   Infraorder <chr>, Family <chr>, Subfamily <chr>, Tribe <chr>, Genus <chr>,
#   comments <chr>, site_full <fct>, log_ave_lit <dbl>

```

c. Code your figure

```

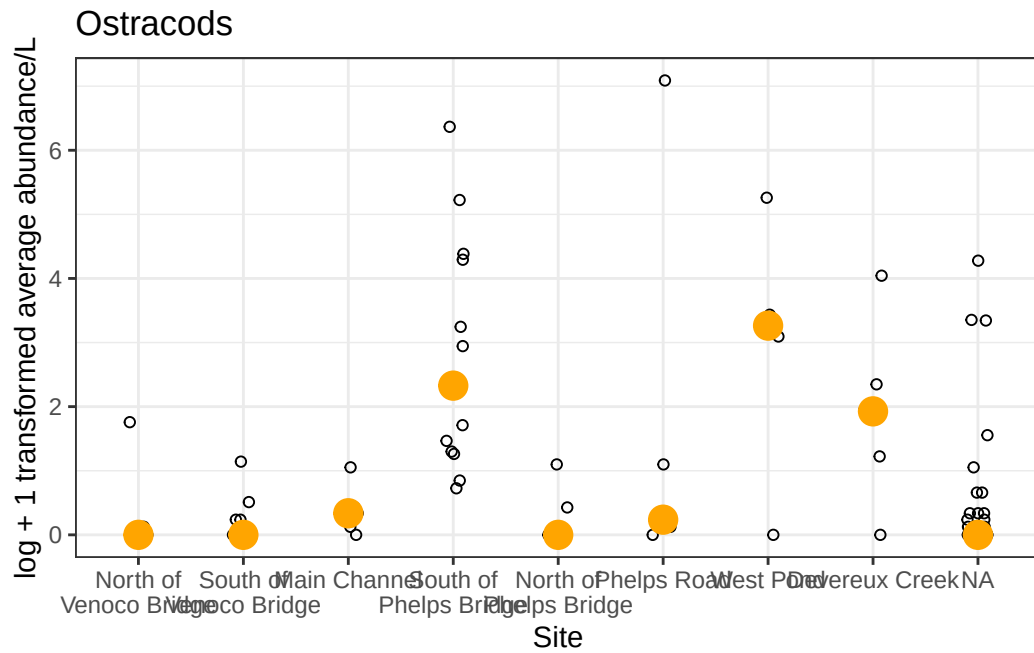
# base layer
ostracodgraph <- ggplot(data = ostracods,
  # x-axis: site
  # y-axis: mean salinity
  mapping = aes(x = site_full,
    y = log_ave_lit)) +
# salinity measurements at each survey
geom_jitter(height = 0,
  width = 0.1,
  shape = 21) +
# wrapping x-axis text
scale_x_discrete(labels = label_wrap(width = 15)) +
# points to represent means
stat_summary(fun = median,
  color = "orange",
  size = 1) +
# labelling x-axis and y-axis
labs(x = "Site",
  y = "log + 1 transformed average abundance/L",
  title = "Ostracods") +

# change theme from default
theme_bw()

# displaying object
ostracodgraph

```

Warning: Removed 9 rows containing missing values or values outside the scale range (``geom_segment()``).



d. Create a multipanel figure

e. Interpret your figure

2. Visualizing total abundance as a function of salinity